OVER A SEPARABLE FRACTION FIELD EXTENSION A SOLUTION CAN BE REPLACED BY A SEPARABLE SOLUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span contains the author's three enumerated hypotheses, delivered verbatim. The author's last sentence begins 'is smaller and we conclude by induction.' as a continuation of the preceding display; that phrasing is reproduced unchanged. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Solutions for an extension of discrete valuation rings and the notion of a separable solution. the hypotheses at statement lines 35897-35899. Fixes the objects being replaced and produced.
- The inseparable degree of a finite extension and its multiplicativity in a tower. used at proof lines 35910-35911 and 35919-35921. Is the induction measure and the computation that decreases it.
- The approximate-solution lemma producing a separable solution after a purely inseparable degree p step. invoked at proof lines 35917-35919. Is the engine of the induction step.

Source reference. Original lemma 030L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3952–3952. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 09HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12006–12006. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 09HK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12008–12008. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 030K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3951–3951. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 35775: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35775–35775. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \subset B$ be an extension of discrete valuation rings. Assume*

(1) *the extension L/K of fraction fields is separable,*
(2) *B is Nagata, and*
(3) *there exists a solution for $A \subset B$.*

Then there exists a separable solution for $A \subset B$.

Proof. The lemma is trivial if the characteristic of K is zero; thus we may and do assume that the characteristic of K is $p > 0$.

Let K_2/K be a solution for $A \to B$. We will use induction on the inseparable degree $[K_2 : K]_i$ (Fields, Definition 030L) of K_2/K. If $[K_2 : K]_i = 1$, then K_2 is separable over K and we are done. If not, then there exists a subfield $K_2/K_1/K$ such that K_2/K_1 is purely inseparable of degree p (Fields, Lemmas 030K and 09HI). By Lemma 35775 there exists a separable extension K_3/K_1 which is a solution for $A \subset B$. Then $[K_3 : K]_i = [K_1 : K]_i = [K_2 : K]_i/p$ (Fields, Lemma 09HK) is smaller and we conclude by induction. □